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PORTFOLIO

Bridging the gap between hardware and software
Cross-Disciplinary Engineer

*Robotics / Computer Vision /
ROS2 / Deep Learning / CFD*

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[GITHUB](#)

[RECOMMENDATION LETTER](#)

Introduction*

SONG JONG JIN

2000.08.31



CONTACT

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FOCUS

Computer Vision (YOLO)
Deep Learning (LSTM)
Isaac Sim
Ansys Fluent

TECHNICAL SKILLS

ROS2
Python / PyTorch
YOLO
Isaac Sim
Ansys Fluent

EDUCATION

B.S. in Mechanical Engineering, University of Ulsan

Expected Graduation: Feb. 2027

Advanced Coursework in Fluid, Aerodynamics & Structural Mechanics

CAREER

Doosan NOKEY Boot Camp

2025.12 - 2026.08

Digital Twin-Based Automation System
Semantic AI-Based Autonomous Drone Assistant
Mechanical Drone-Based Source Control System
Museum Security Monitoring & Multi-Drone Tracking System

Undergraduate Researcher – Transitional Turbulence Flow Lab

2023.03 - 2024.01

Propeller Drone Fluid Analysis
Nozzle Combustion Analysis

Projects*

Doosan ROKEY Boot Camp

04 PROJECTS

01 **Isaac Sim-Based Autonomous Navigation System**
Isaac Sim, Reinforcement Learning, ROS2 Nav

02 **Semantic AI-Based Autonomous Assistant**
Local LLM, VLA

03 **M0609-Based Braille Production System**
Integrated Node, Robot Motion Code

04 **Museum Security Monitoring & Tracking System**
ROS2, YOLO

Project Introduction*

Isaac Sim-Based Autonomous Navigation System



Project Overview | Duration: 2 weeks | Team: 5

KEY TECHNOLOGIES

Doosan M0609
Ros2
Isaac Sim
Local LLM

MY RESPONSIBILITIES

Implemented autonomous navigation using ROS2 Nav
Implemented reinforcement learning for posture recovery in Isaac Sim
Implemented autonomous operation using a local LLM

My Role

- FLORENCE-2
- Task-Oriented LLM
(Reasoning, Questions)
- Coder LLM
(Code Generation, Execution)

Dual-LLM approach

-Task-oriented LLM issues detailed commands

Command Execution Process:

LLM autonomously selects modules required for the task
Modules are then executed sequentially

Details

- 01 PICK
- 02 PLACE
- 03 MOVE
Used 3D LiDAR to determine current position and navigate autonomously
Implemented obstacle avoidance
- 04 Posture Recovery
Estimated posture using an IMU sensor
Recovered posture using the reinforcement learning policy

Project Introduction*

Isaac Sim-Based Autonomous Navigation System

Code LLM Performance Comparison & Analysis

Model	Time Taken	Tasks Generated
Qwen3.5:4b	37.036s	6 steps
Qwen3.5:9b	66.893s	7 steps
Qwen3-coder:30b ★	11.069s	7 steps

✓ Selected the model that generated faster, more detailed tasks:
Qwen3-coder:30b

Issues with Qwen3-Coder:30B

⚠ **Hardware Constraints & Memory Leakage**

QQwen-Coder:30B requires at least 18 GB of VRAM

Rtx 50RTX 5080 provides ~15 GB usable VRAM, causing memory overflow

When memory is saturated, AI inference speed and model performance drop sharply

🔧 **Solution & Optimization**

Forced garbage collection before each AI operation

RESULT : RESULT: Consistent and stable output quality

Project Introduction*

Isaac Sim-Based Autonomous Navigation System

01 Nav2 Navigation Precision Optimization

Problem

Sensor/map mismatch and target-position error

Solution

Adjusted Nav2 parameters and initial pose

Shortened sensor data update intervals

Improved navigation precision by tuning goal tolerance and initial pose

LiDAR Update Interval

0.25 m → 0.05 m

0.20 rad → 0.02 rad

Goal Position Error

0.25 m → 0.05 m

initial_pose Y Correction

RESULT : Improved 2D LiDAR visualization and Grid Mapping alignment

02 RL-Based Self-Recovery

Problem

Failure to recover posture after tipping over

During training, the policy acted immediately after tipping

Whereas in Isaac Sim, it acted only after fully falling

This small difference caused recovery failure



Solution

Redesigned environment and reward structure

Reset the environment to recover only after fully falling

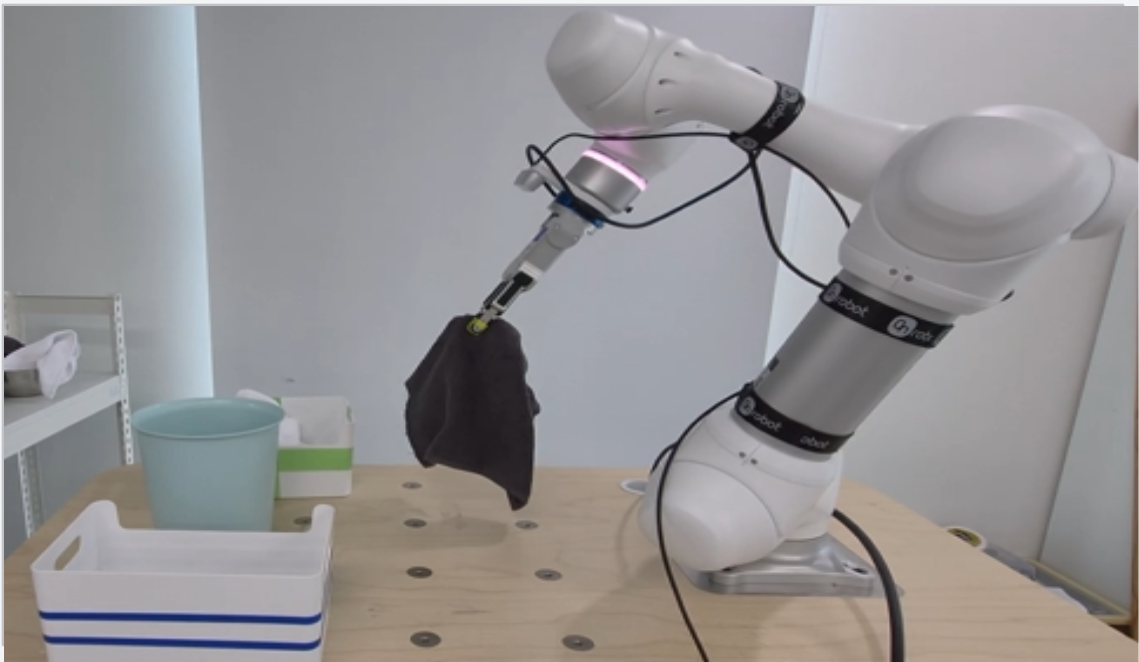
Redesigned rewards to minimize arm movement



RESULT : with only 1/6 the previous iterations successfully stood upright

Project Introduction*

Semantic AI-Based Autonomous Drone Assistant



Project Overview | Duration: 2 weeks | Team: 5

KEY TECHNOLOGIES

Doosan M0609
Ros2
YOLO
Local LLM

MY RESPONSIBILITIES

Implemented VLA using Qwen and Llama
Implemented a robot that recognizes hand gestures and executes commands

My Role

Web + Data Base

Task-Oriented LLM
(Reasoning, Questions)

Coder LLM
(Code Generation, Execution)

Dual-LLM approach

- Task-oriented LLM issues detailed commands
- Code generation vs. module selection

Command Execution Process:

Generate code when a code-generation command is issued
Execute the corresponding module when a module command is issued

Details

01 YOLO

Generated contours to guide grasping toward the highest point

02 Code Generation

Implemented autonomous LLM-based Pick & Place code generation logic

03 Exception Handling

Implemented detailed M0609 actions based on finger counts

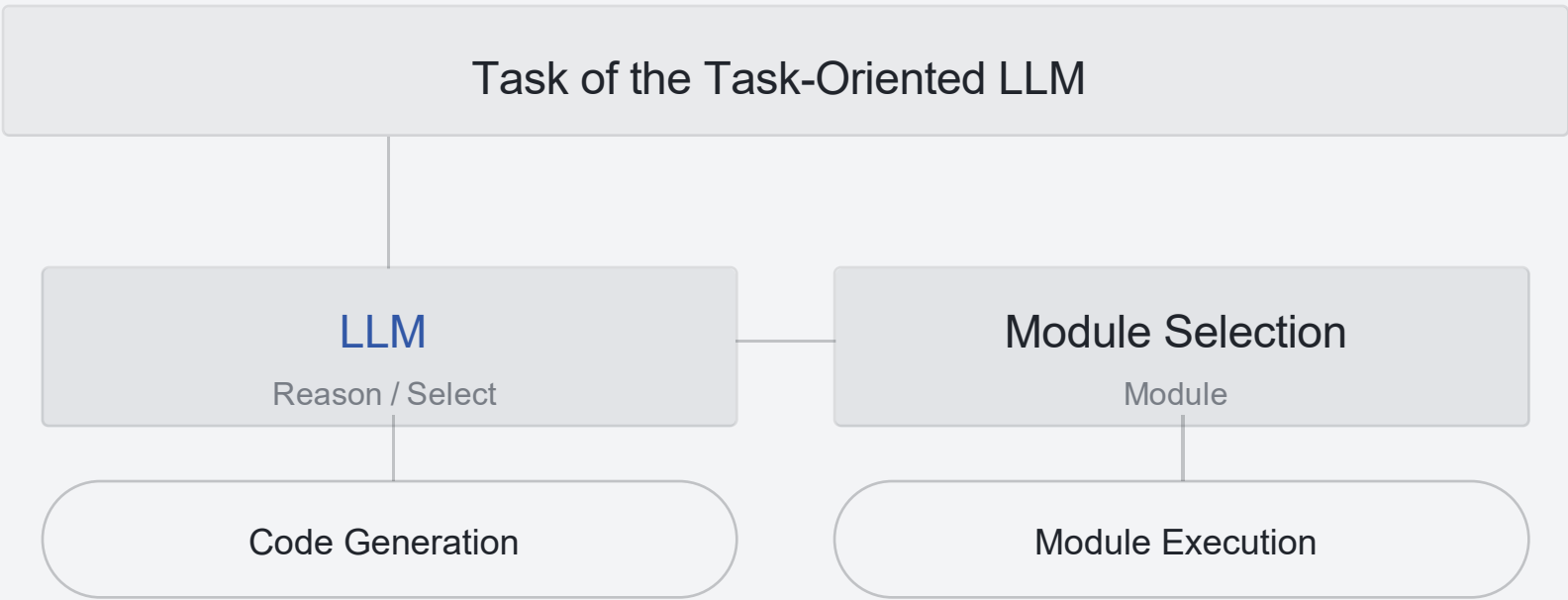
- 1 finger: X-axis rotation
- 2 fingers: Return to initial state
- 3 fingers: Pick & Place
- 4 fingers: Detect pointing direction
- 5 fingers: Track palm

Semantic AI-Based Autonomous Drone Assistant

Code LLM Development Process

Final Control Architecture

Reduced the LLM workload and modularized standardized actions



Generate code only for core actions; call API modules for repetitive actions

❏ Existing Approach: Full-code generation with a local LLM

Required complex prompts with strict constraint conditions
Long prompt processing delayed code generation (~34 s)
Needed to reduce physical execution time while maintaining control performance

⚠ Improvement 1: Local LLM Masking Architecture

Designed a multi-step LLM pipeline to reduce prompt dependency
Small-parameter LLM failed to generate precise control code due to physical limitations
→ Limitation: High performance was difficult within limited resources

❏ Improvement 2 (A): Prompt Lightweighting & Optimization

Switched to English-based templates to maximize inference performance
Compressed tokens by minimizing natural-language descriptions and focusing on API specifications
→ Result: 1/3 shorter generation time and 80% prompt reduction (real-time performance achieved)

⚙ Improvement 2 (B): Hybrid Modular Architecture

Generated code only for core tasks (Pick & Place); modularized other functions
Autonomous branching: direct code generation vs. predefined API module calls
→ Result: System autonomy and execution-code reliability maximized

Project Introduction*

Semantic AI-Based Autonomous Drone Assistant

▣ Fallback Scenario for YOLO Detection Failure

Implemented a fallback control logic that tracks the pointing direction by finger-count mapping

⚠ Key Limitations & Issues

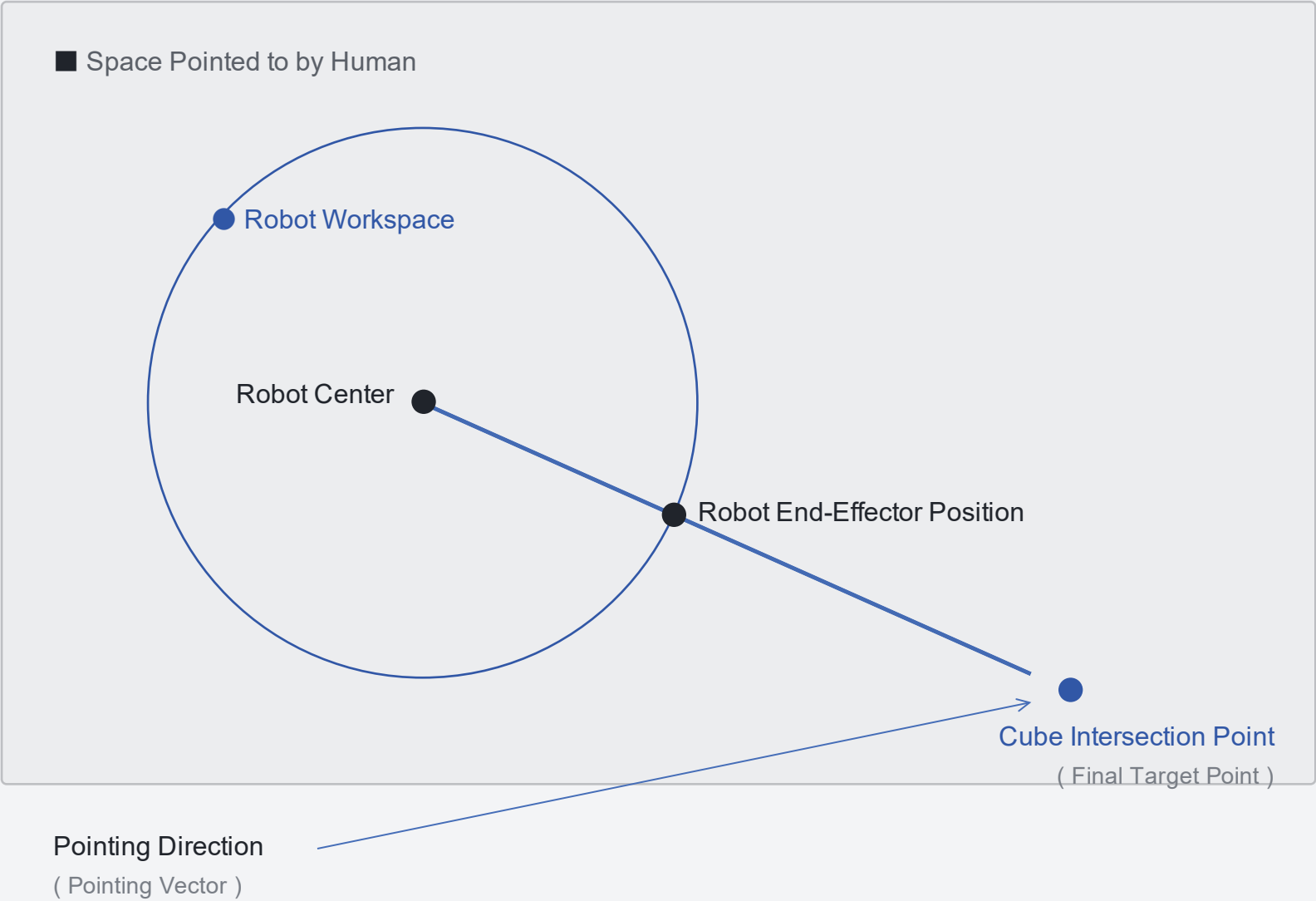
- 1. Precise coordinates of the pointing direction could not be determined
- 2. No intuitive logic for directing the robot after coordinate detection

✓ Solution Through Geometric Mapping

Designed a virtual cuboid: calculated the intersection with the pointing vector to derive the final target point

Virtual spherical control: defined the robot workspace as a sphere for stable fingertip coordinates

Geometric Control Mapping Model



Project Introduction*

Braille Business Card Embossing System



Project Overview | Duration: 2 weeks | Team: 5

KEY TECHNOLOGIES

YOLO
ROS2
Doosan M0609

MY RESPONSIBILITIES

Developed Braille conversion logic
Developed Braille embossing algorithm
System Integration

My Role

Web + Data Base

Braille Conversion:
Convert Korean text input to Braille

Doosan M0609

Braille Embossing:
Emboss Braille stored in the database

Integrated Node:
Integrated each node using ROS2 Actions

Exception Handling:
Implemented recovery algorithm
for emergency stops

Details

01 Braille Conversion
1-to-1 Braille conversion using the JAMO library
Added exception rules from the National Institute of Korean Language

02 Braille Embossing
Decomposed forces using inverse kinematics
Fine error correction

03 Integrated Processing
Built an integrated node for five individual nodes and exception handling

04 Exception Handling
Restarted the system after recovery from an emergency stop

Project Introduction*

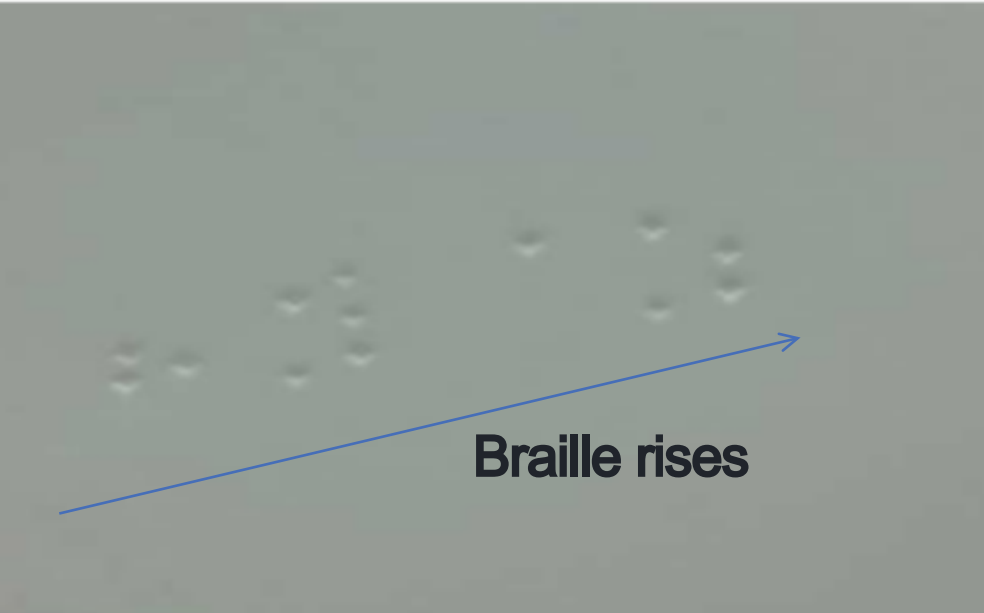
Braille Production

⚠ Problem Identified

Tool tilting or
slight misalignment during embossing

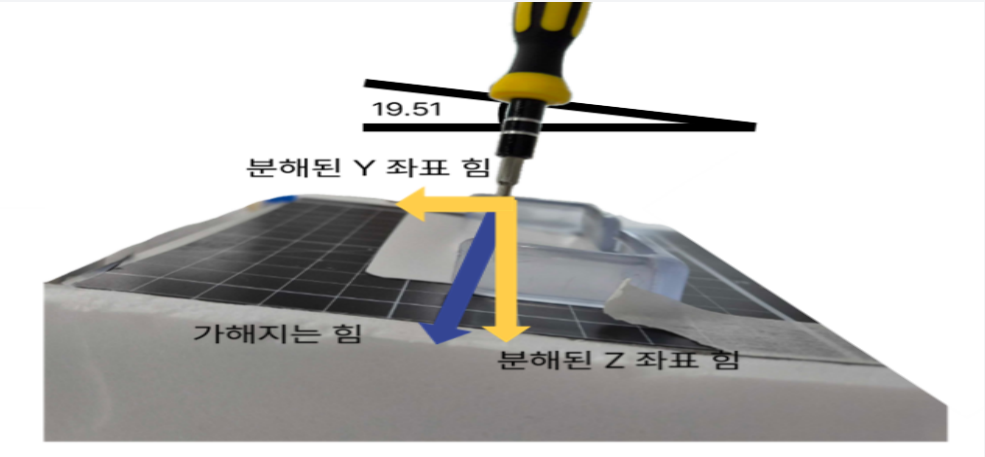
Braille damage occurred

The Braille verification algorithm also reported a mismatch



□ Root Cause Analysis & Solution

- 01** Embossing Direction Correction
Analyzed the cause of force-direction misalignment and adjusted the robot-arm coordinate axes based on force decomposition
- 02** Mechanical Deviation Correction
Identified small positional deviations during embossing and applied offsets using simulation-based correction logic



Embossing Plate Tilt / Gripping Deviation

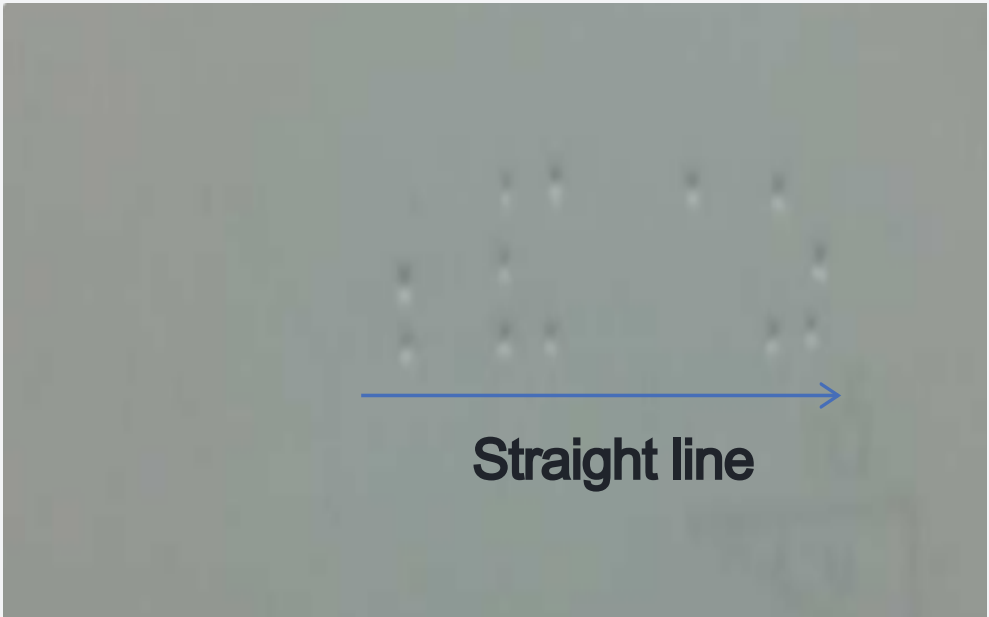
→ Correction Logic →

RESULT : Corrected Embossing Posture and Position

✓ Final Result

Verification based on NIKL standard rules

Verified the adjusted embossing result with the algorithm and obtained the final “Match (Normal)” result



RESULT

Improved embossing precision and completed the Braille business card production process

Project Introduction*

Braille Production

Integrated Module Structure

Individual Nodes by Function

Node 01	Pick Paper
Node 02	Embossing Control
Node 03	Flip Paper
Node 04	Stamp
Node 05	Complete

+

Exception Handling

Integrated Control Node

Managed execution order and status of each functional node using ROS2 Actions

⚠ Bottleneck Occurred

Problem

A bottleneck prevented transition to the next action during integration

Solution

Separated Communication Tasks

FireBase Separated Firebase communication into a multithreaded process
Ros2 Executor Prevented ROS2 Executor blocking

Variable Step introduced

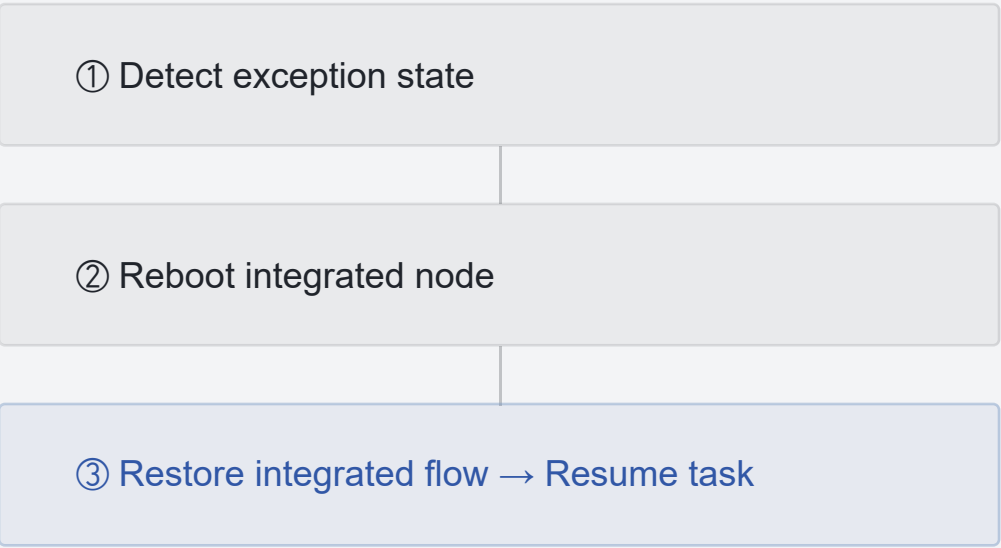
Result for the next action Send goinstead of Stepused a variable to manage state

Exception Handling & Recovery

Problem

During restart after an emergency stop, zombie nodes occurred during action execution

Solution



Goal: Recover without interrupting the overall workflow

Project Introduction*

Museum Security Monitoring & Tracking System



Project Overview | Duration: 2 weeks | Team: 9

KEY TECHNOLOGIES

YOLO
ROS2
OAK-D Camera

MY RESPONSIBILITIES

Trained on images of objects, security guards, etc.
Calculated distance using a depth camera
Developed inter-robot collision avoidance logic

My Role

TURTLE BOT

YOLO + Camera

During Patrol:

Detect museum artifacts using YOLO

When Anomaly Occurs:

Notify the web when an artifact is missing
Send TurtleBot dispatch signal

When Intruder Is Detected:

Send intruder location to TurtleBot
Calculated distance to TurtleBot using a depth camera

Web + Data Base

Details

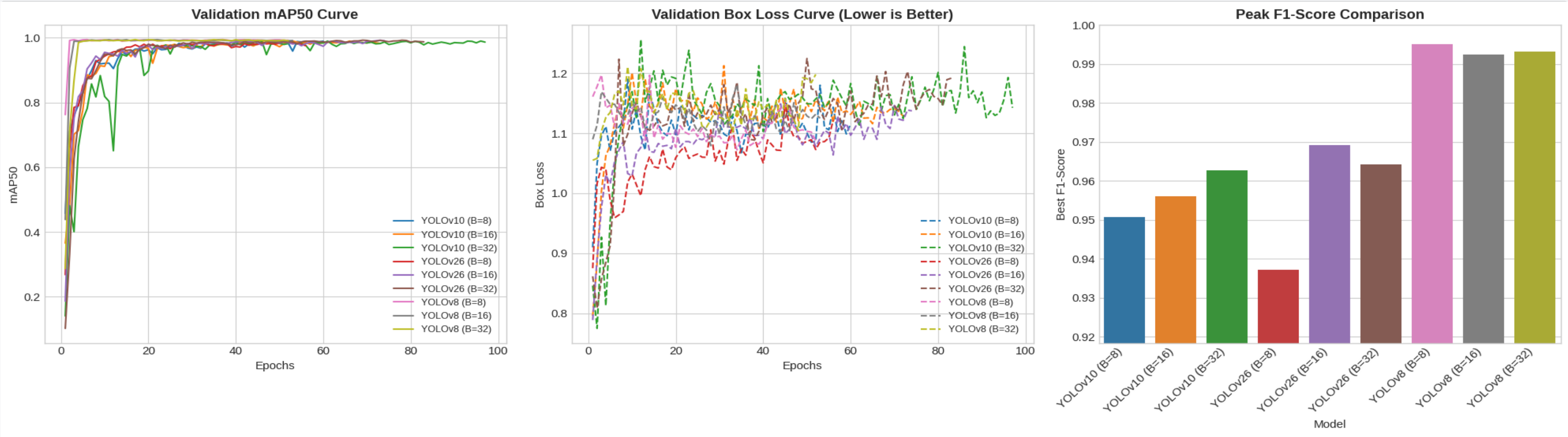
01 Developed collision avoidance algorithm
Developed an encirclement algorithm using two TurtleBots

02 YOLO Object Detection
YOLO fine-tuning
Compared detections with the database and implemented anomaly-alert logic

03 Distance Calculation
Implemented target-distance calculation logic

Project Introduction*

Museum Security Monitoring & Tracking System



YOLO Model Training

Target model versions: YOLO V8, V10, V26

Compared mAP and F1-Score by batch

Selected Best-Performing Model

YOLO 8 achieved a high mAP50 score

Batch 8 achieved a high F1-Score

RESULT : RESULT: Selected YOLO v8, Batch 8

Project Introduction*

Museum Security Monitoring & Tracking System

✗ Problems with Existing Approach

01 Simultaneous Detection

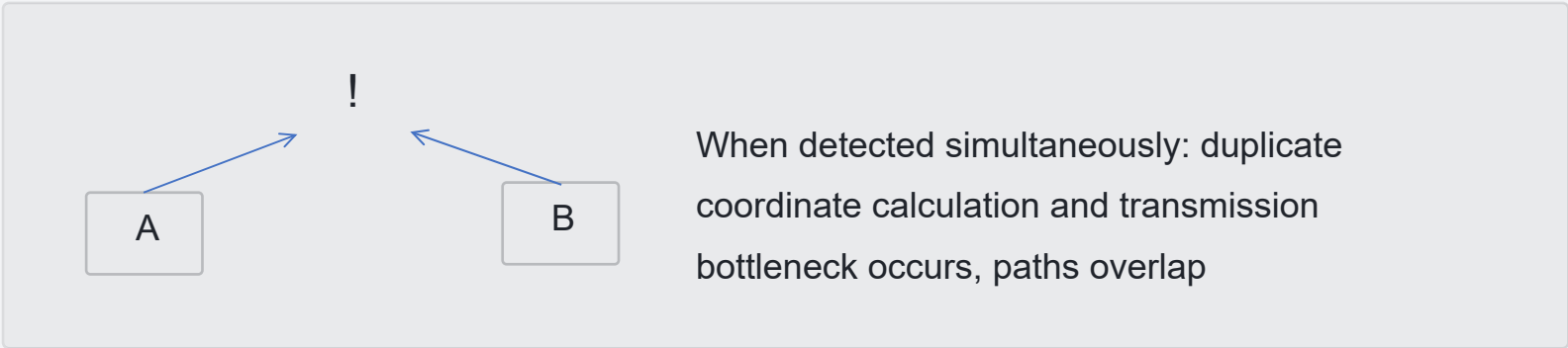
Two TurtleBots detect the same intruder simultaneously

02 Duplicate Computation & Transmission

Each robot independently calculates and transmits coordinates over the network

03 Bottleneck & Collision Risk

Simultaneous communication creates network bottlenecks
Overlapping paths to the same target increase collision risk



✓ Improved Approach: Sequential Relay via a Lead Robot

01 Assign Lead Robot

The first robot to detect becomes the lead robot

02 Lead Coordinate Computation

Only the lead robot calculates the intruder's representative coordinates

03 Relay to Following Robot

Safely sends relative coordinates to the following robot

04 Collaborative Tracking

inter-robot spacing and tracks while considering relative positions

RESULT : 1 lead computes + 1 receives

*Retrospective**

Doosan ROKEY Boot Camp Retrospective

01 Value of Simulation

By applying Nav2 and SLAM used in the real environment directly to Isaac Sim, I recognized the value of simulation. Reproducing autonomous navigation in a virtual environment and repeatedly validating diverse scenarios showed me that simulation is an essential environment for validating technology and identifying problems before real-robot development.

02 Designing the Role of AI

I experienced the limitations of having AI handle everything directly. By separating standardized actions into modules and focusing the LLM on code generation, I learned that defining AI's role around the actual task and designing the system accordingly is more important than using a general-purpose AI as-is.

03 Importance of Integration

I experienced that unexpected errors can occur during integration even when individual functions work correctly. When code conflicts delayed the Braille production project, I resolved the issue by reallocating team members. This taught me the importance of directly experiencing various failures and responding appropriately during system development.

“From the value of simulation to the role of AI and the importance of system integration, I broadened my perspective on robot development.”

Projects*

Undergraduate Researcher

02 PROJECTS

01

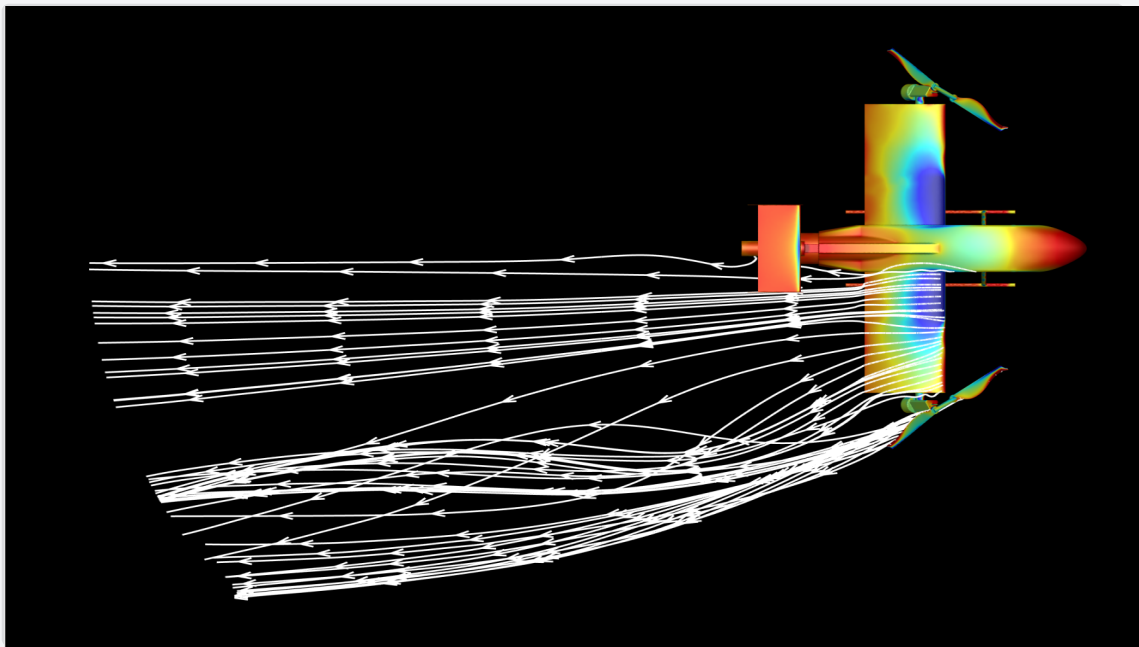
Variable-Pitch Drone Propeller CFD Analysis
ANSYS Fluent, CATIA V5

02

Rocket Nozzle Design & CFD Analysis
ANSYS Fluent, CATIA V5

Project Introduction*

Variable-Pitch Propeller Drone CFD Analysis



Project Overview | Duration: 2 months | Team: 4

KEY TECHNOLOGIES

Ansys Fluent
CATIA V5

MY RESPONSIBILITIES

Mesh Generation
Fluent Simulation
Contour Report

My Role

3D Model Modification

Modified the 3D model to facilitate mesh generation

Ansys Fluent

Mesh Generation:

Generated a 9-million-element mesh

Flow Analysis:

Calculated pressure, velocity, and streamlines
Derived thrust and lift

Details

01 Propeller Tip Modification

Added rounding to the propeller tip for mesh quality

02 Mesh Quality

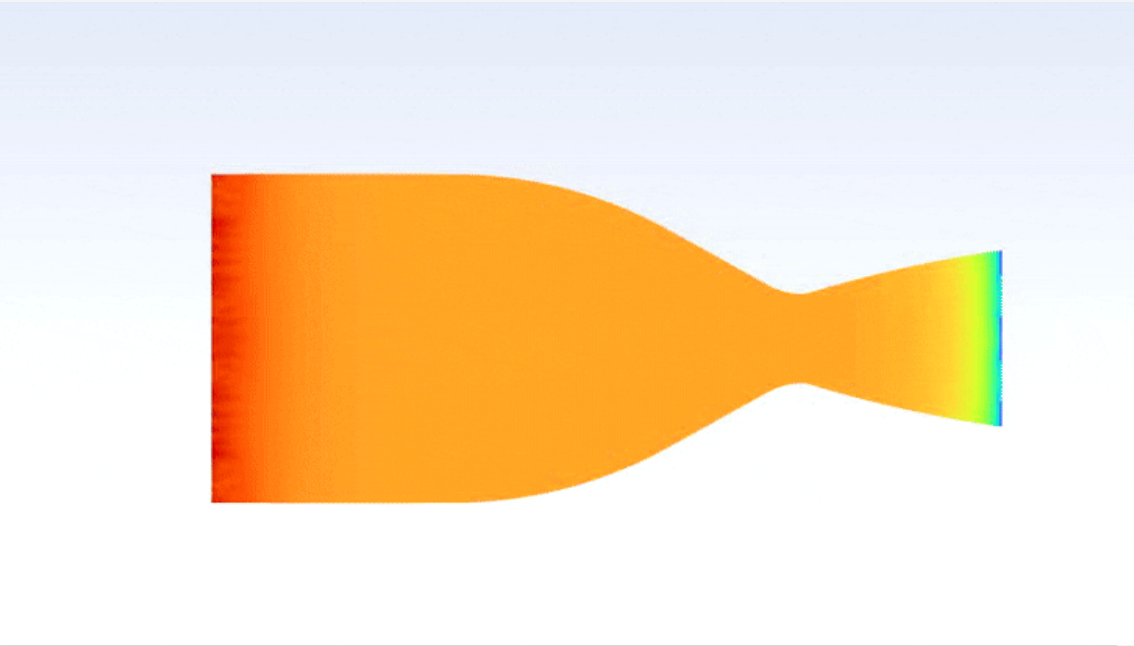
Kept skewness below 0.9
Set $Y^+ = 2.61$
Generated and compared four meshes with different element sizes to select the optimum mesh

03 MRF Method

Reduced computation time using steady-state calculations

Project Introduction*

Rocket Nozzle Design & CFD Analysis



Project Overview | Duration: 2 months | Team: 1

KEY TECHNOLOGIES

CATIA V5
ANSYS FLUENT

MY RESPONSIBILITIES

2D & 3D Geometry Modeling
Added combustion reaction equations
Transient Flow Analysis

My Role

Geometry Modeling

Designed the nozzle using CATIA V5

Ansys Fluent

Combustion Reaction:

Added kerosene–oxygen chemical reaction equations

Transient State:

Calculated the combustion process over time

Analyzed the cause of 3D geometry calculation failure

Details

01 Bell Nozzle

Designed a bell nozzle using calculations to achieve the target thrust

02 Combustion Reaction

Added reaction equations to calculate pressure and thrust according to fuel quantity

03 Simulation

Simulated the process from ignition until the flow exits the nozzle

04 Fuel Backflow

Identified backflow caused by excessive internal pressure after combustion

*Retrospective**

Undergraduate Research Retrospective

01 Importance of Simulation Conditions

I found that even the same geometry can produce slightly different results depending on the mesh and boundary conditions. This taught me to understand why each condition is applied when setting up a simulation.

02 Understanding Physical Phenomena

By analyzing drone propellers and rocket nozzles, I learned that results can only be interpreted correctly by understanding physical phenomena such as flow, pressure, and temperature - not just numbers and graphs.

03 Validation of Simulation Results

When results differed from expectations, I repeatedly checked the mesh, boundary conditions, and simulation model rather than simply accepting the output. This developed my habit of critically reviewing simulation results..

“I learned to think like an engineer by examining not only simulation results, but also the conditions and physical causes behind them.”

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Thank you

Robotics / Computer Vision / ROS2 / Deep Learning / CFD

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RECOMMENDATION LETTER https://drive.google.com/file/d/1CpsjVO26XAU-IS09odG9ScP_uCqnZtph/view